

■■ MoodMeal — Project Enhancement & Optimization Plan

■ Part 1 — Real-World Expansion Ideas

- ■ Mental Wellness Assistant — Suggest comfort foods and mindfulness tips for emotional balance.
- ■ Diet-Aware Recommender — Add user profiles (age, allergies, calorie goal) for healthy, personalized suggestions.
- ■ Cultural Cuisine Mapper — Link emotions with regional cuisines for diverse comfort recommendations.

■■ Part 2 — Technical Enhancements for Efficiency

- Use smaller models like DistilBERT-mini for faster emotion detection.
- Cache models using `@st.cache\_resource` to prevent reloading each chat.
- Add local caching for images and use Unsplash API for high-quality, consistent visuals.
- Introduce conversation context memory to reduce misclassification of neutral inputs.

■ Part 3 — UI & UX Enhancements

- Add gradient header and modern layout using Streamlit custom HTML/CSS.
- Display emotion icons (■ ■ ■) next to detected emotion.
- Include analytics tab for emotion/food trends using bar charts.
- Enable voice input/output via `speech\_recognition` and `gTTS` for accessibility.

■ Part 4 — Intelligence Improvements

- Integrate VADER or TextBlob for emotion intensity detection.
- Recommend foods based on time of day + emotion (e.g., 'sad' + 'night' → warm soup).
- Add user feedback tracking to refine recommendations.
- Introduce multilingual support using Google Translate API.

■ Part 5 — Research & Future Scope

- Focus research title: 'MoodMeal: A Hybrid AI Chatbot for Emotion-Driven Dietary Recommendation'.
- Integrate nutritional APIs (Spoonacular, Edamam) for healthier diet advice.
- Extend to wearable integration (Fitbit, Google Fit) for real-time emotional health tracking.
- Explore emotion trend prediction over multiple sessions.

■ Optional — Backend Optimization

- Wrap NLP pipelines into asynchronous tasks or a FastAPI microservice to reduce Streamlit latency.
- Enable GPU runtime for heavy model inference tasks if running on Colab or local CUDA setup.

■ TL;DR — Top 5 Upgrades

- ■ Improve Emotion Detection Accuracy and Cache Results
- ■■ Add Time/Diet-Aware Food Mapping
- ■■ Replace API with Curated/Unsplash Image Sources
- ■ Build Interactive Streamlit UI with Icons and Charts
- ■ Maintain Context Memory per Session for Natural Conversation